Jakub Nawała

Curriculum vitae

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About Me

I'm a Research Engineer with more than 10 years of experience specialising in Video Quality Assessment, Computer Vision, and Video Compression. Over the years I've participated in more than 10 R&D projects working with both academic and industrial partners. As of 2026, I have published 25 technical papers, some of which were accepted at top-tier conferences, including the European Conference on Computer Vision and ACM International Conference on Multimedia.

Technologies and Languages

Python	7 years of experience	**Matlab**	8 years of experience
C/C++	7 years of experience	**Java**	Basic familiarity
DBS	MySQL, MongoDB	**GUI Design**	Qt
R	Basic familiarity		
Embedded	NVIDIA Jetson, ARM Cortex-M0+/M3		
Sys Admin	Linux, SSH, Git, SVN, Bash, ZSH, Batch, PowerShell		
Video Compression	H.264, H.265, AV1, AV2, Deep-learning based coding tools (PP & SR)		
Multimedia Processing	FFmpeg, VLC, OpenCV, Pandas, NumPy, scikit-video, CUDA, ZMQ		

Work Experience

Aug. 2024 – Present **Research Associate in Audience Understanding**,
University of Bristol, Bristol, UK.
- Spearheaded experimental design of a novel research work on thermal imaging in the context of audience understanding.
- Implemented a multi-tech stack image analysis pipeline for a thermal camera.
- Worked on a multidisciplinary team made up of neuroscientists, psychologists, engineers, and creatives.
- Collected and analysed complex, real-world datasets of audience engagement data.

Sept. 2022 – July 2024 **Research Associate in Perceptually Optimised Video Compression**, *University of Bristol*, Bristol, UK.
- Conducted project-based applied research in collaboration with Netflix engineers.
- Implemented a video data pipeline in Python that operates on TBs of raw video data using a large-scale distributed system.
- Integrated machine-learning models (implemented in PyTorch) with the C code of the AVM video codec.
- Helped develop neural networks for image super-resolution and post-processing.
- Optimised a deep learning solution in terms of computing and memory footprint.
- Authored technical papers in the field of AI-based video compression.

Oct. 2020 – June 2022 **PhD Student Researcher**, *NTNU & AGH UST*, Kraków, Poland.
- Implemented Python scripts processing a huge dataset with millions of entries.
- Using a parallel/distributed computing platform to speed up processing of huge data sets.
- Coordinated a small team of researchers.
- Worked in the Agile methodology.

Mar. 2019 – July 2020 **PhD Student Researcher**, *Huawei Technologies Duesseldorf GmbH & AGH UST*, Kraków, Poland.
- Helped develop a machine learning-based solution assessing image quality in the context of computer vision tasks.
- Implemented Python scripts simulating various video capture distortions.
- Helped evaluate various image segmentation and image classification computer vision algorithms in the context of automatic licence plate recognition, autonomous driving, face detection, and face recognition.
- Contributed to exploratory data analysis and feature engineering in the context of computer vision algorithms.
- Used Git to collaborate with other team members.

Sept. 2018 – Aug. 2019 **PhD Student Researcher**, *Netflix, Inc. & AGH UST*, Kraków, Poland.
- Implemented Python scripts running data processing and analysis of huge data matrices with millions of entries.
- Used a parallel/distributed computing system to speed up statistical modelling required for the project.
- Implemented data visualisation techniques.
- Lead data cleansing effort.

Mar. 2018 – Dec. 2018 **PhD Student Researcher**, *Net Research sp. z o.o. & AGH UST*, Kraków, Poland.
- Co-authored a successful funding bid to transfer IP from an academic team into an innovative IPTV product.
- Implemented Python scripts detecting lip sync issues in audio-visual sequences using computer vision algorithms (including image classification and facial landmark detection).
- Implemented in C/C++ a custom VLC plugin.

Mar. 2017 – June 2019 **Video Quality Technical Expert**, *Net Research sp. z o.o.*, Kraków, Poland.
- Implemented in C/C++ a GUI in Qt for video quality analysis of IPTV streams.
- Implemented in Python video quality analysis algorithms.
- Used Git and SVN to collaborate with other team members.
- Worked with MongoDB as the database solution for a GUI app.

Dec. 2015 – Mar. 2017 **Student Researcher**, *AGH UST*, Kraków, Poland.
- Implemented in C/C++ CUDA kernels on the NVIDIA Jetson platform in the context of underwater image segmentation.
- Used Git to collaborate with other team members.

Feb. 2015 – Sept. 2016 **Student Researcher**, *AGH UST & 11 international partners*, Kraków, Poland.
- Implemented in Python software injecting visual artifacts into videos.
- Used Git to collaborate with other team members.

Feb. 2015 – Nov. 2015 **Student Researcher**, *AGH UST & 3 international partners*, Kraków, Poland.
- Implemented in C/C++ and Python video quality analysis algorithms.
- Used Git to collaborate with other team members.

Selected Publications

[1] Y. Jiang, J. Nawała *et al.*, "Rtsr: A real-time super-resolution model for av1 compressed content," in *2025 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2025, pp. 1–5.

[2] J. Nawała, Y. Jiang, F. Zhang, X. Zhu, J. Sole, and D. Bull, "Bvi-aom: A new training dataset for deep video compression optimization," in *2024 IEEE International Conference on Visual Communications and Image Processing (VCIP)*, 2024.

[3] Y. Jiang, J. Nawała, F. Zhang, and D. Bull, "Compressing deep image super-resolution models," in *2024 Picture Coding Symposium (PCS)*, 2024, pp. 1–5.

[4] B. Ćmiel, J. Nawała *et al.*, "Generalised score distribution: Underdispersed continuation of the beta-binomial distribution," *Statistical Papers*, 2023.

[5] L. Janowski, J. Nawała, T. Hoßfeld, and M. Seufert, "Experiment precision measures and methods for experiment comparisons," in *2023 15th International Conference on Quality of Multimedia Experience (QoMEX)*, 2023, pp. 49–54.

[6] J. Nawała, L. Janowski *et al.*, "Generalized score distribution: A two-parameter discrete distribution accurately describing responses from quality of experience subjective experiments," *IEEE Trans. Multimedia*, vol. 25, pp. 6090–6104, 2023.

[7] J. Nawała, L. Janowski, B. Ćmiel *et al.*, "Reproducibility companion paper: Describing subjective experiment consistency by p-value p–p plot," in *Proceedings of the 29th ACM International Conference on Multimedia*. ACM, 2021, pp. 3627–3629.

[8] J. Nawała, L. Janowski, B. Ćmiel, and K. Rusek, "Describing subjective experiment consistency by p-value p-p plot," in *Proceedings of the 28th ACM International Conference on Multimedia*. ACM, 2020, pp. 852–861.

[9] J. Nawała, M. H. Pinson, M. Leszczuk, and L. Janowski, "Study of subjective data integrity for image quality data sets with consumer camera content," *Journal of Imaging*, vol. 6, pp. 1–21, 2020.

Research Interests

Statistical Modelling, Computer Vision, Video Compression, Multimedia Quality Assessment, Quality of Experience, Perceptual Video Quality Assessment, Subjective Testing, Subjective Data Analysis.

Research Activity

2017 – Present **Video Quality Experts Group (VQEG)**, Statistical Analysis Methods group.
Among others, made a contribution to ITU-T Rec. P.910 (10/23).

2015 – 2022 **AGH Video Quality of Experience Team**.
https://qoe.agh.edu.pl

Education

2018 – 2023 **PhD** w. honours, *Information and Communication Technology*, AGH UST, Kraków, Poland.
Thesis: "Modelling the Response Generation Process in Multimedia Quality of Experience Assessment Subjective Experiments."

2017 – 2018 **M.E.** w. honours, *Electronics and Telecommunications*, AGH UST, Kraków, Poland.
Thesis: "Developing a Quality Indicator Detecting Compression Blur Distortion."

2013 – 2017 **B.E.** w. honours, *Electronics and Telecommunications*, AGH UST, Kraków, Poland.
Thesis: "GPU Accelerated Algorithms for Image Enhancement and Filtering in Real-time."